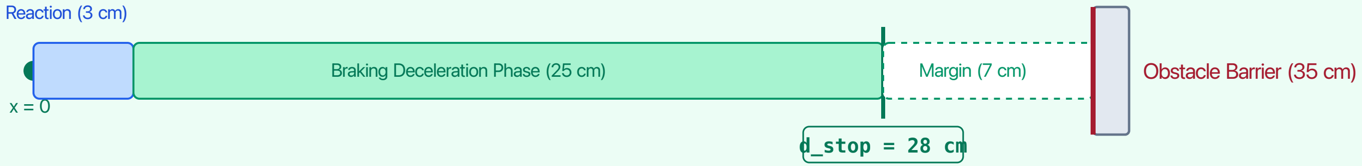
DYNAMIC STOPPING DISTANCE PHYSICS (d_stop)

Translating milliseconds of computational delay into physical centimeters of collision hazard: $d_{stop} = v \cdot \Delta t_{delay} + v^2 / (2 a_{max})$

Governing Physics: Total Stopping Distance = Reaction Travel ($v \cdot \Delta t$) + Braking Travel ($v^2 / 2a_{max}$) | Parameters: $v = 1.0$ m/s, $a_{max} = 2.0$ m/s², Clearance Barrier = 35 cm

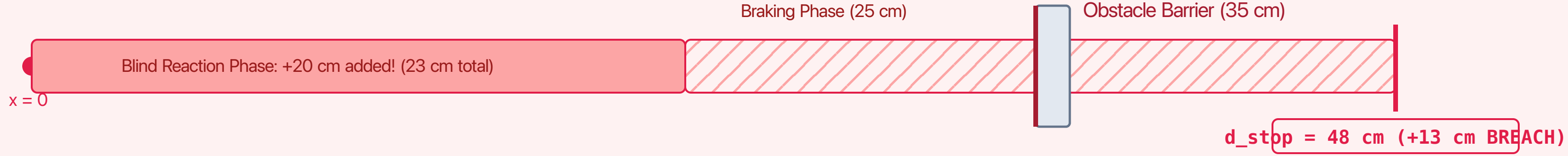
SCENARIO A · NOMINAL P50 LATENCY (t_delay = 30 ms)

VERDICT: SAFE STOP (+7 cm margin)



SCENARIO B · P99 LATENCY TAIL SPIKE (t_delay = 230 ms · OS Jitter + UMA Contention)

VERDICT: SEVERE COLLISION (13 cm breach)



The Invariant Enforcement Mandate: Because an untrusted foundation model on the MPU cannot guarantee bounded latency tails under UMA memory contention, the dedicated 1 kHz MCU must continuously evaluate *d_stop(v) ≤ d_clearance* and preemptively initiate safe braking.